Dynamic Report

About this app

This app was developed to be a user-oriented, open-source environmental DNA analysis tool. It is meant to distill information into helpful graphics and analyses that can be used for management reports. This report is a distillation of the analyses ran within the app. If anything is missing, feel free to go back into the app to perform those analyses and then rerun this report.

This was developed with support from the Marine Conservation Lab at the Smithsonian Environmental Research Center.



Dataset overview

This section will go over the characteristics of the dataset that you analyzed. *This is general information that will speak to the quality of the data and how to interpret the results.*

This section will review the blanks you included, if any, including how many sequences each blank had and if any of them were taxonomically assigned. It will also report whether you included any unassigned sequences in the subsequent analysis, and what normalization method you used, if any.

In addition, this will summarize some basic features of the data, including the distribution of sequences per sample in the total dataset and the one you ultimately analyzed.

About the data manipulation choices:

These samples were selected as your blanks:

```
## [1] "NA"
```

You selected these filtering and normalization methods:
Were unassigned sequences removed? NA
Were blanks removed? NA
Normalization method: NA

```
## Error in incProgress(1/5): 'session' is not a ShinySession object.
```

About your data:

Distribution of original data:

```
## [1] "Characteristics of your sequencing run were not calculated"
```

Number of sequences analyzed:

```
## [1] "Characteristics of the analyzed dataset were not calculated"
```

Total number of sequences: NA Number of assigned sequences: NA

Total number of OTUs: NA Number of assigned OTUs: NA

Blanks

Number of sequences in blanks: NA Number of OTUs in blanks: NA

```
## Error in incProgress(1/5): 'session' is not a ShinySession object.
```

Descriptive statistics

Sequence distribution:
In this plot, we are looking at the number of sequences in each sample. The horizontal line shown is a general marker to show the target minimum number of sequences per sample.
This tool is a useful gut check to get a high-level overview of the entire dataset. Do any samples have significantly more or less sequences than another? That is normally a signal that some kind of normalization will be needed prior to performing any analyses.

```
## Error in plot_bar(params$ps, title = "Number of sequences per sample as compared to the goal sequenc
```

Number of unique taxa:

```
## [1] "Taxonomic diversity was not summarized"
```

Percent composition - how much of the total did each taxa represent?

```
## [1] "NA"
```

```
## Error in incProgress(1/5): 'session' is not a ShinySession object.
```

Alpha diversity plot

Diversity metric used: NA

```
## [1] "NA"
```

```
## Error in incProgress(1/5): 'session' is not a ShinySession object.
```

Ordination plot

Ordination method: NA
Type: NA

```
## [1] "NA"
```

```
## Error in incProgress(1/5): 'session' is not a ShinySession object.
```

R Session information

R packages used to generate the information in this report

```
## R version 4.2.3 (2023-03-15)
## Platform: x86_64-apple-darwin17.0 (64-bit)
## Running under: macOS Big Sur ... 10.16
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.2/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.2/Resources/lib/libRlapack.dylib
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
## [1] kableExtra_1.4.0 knitr_1.45       shiny_1.9.1      ggplot2_3.5.1
##
## loaded via a namespace (and not attached):
##  [1] Rcpp_1.0.11       pillar_1.10.1     compiler_4.2.3    later_1.3.2
##  [5] tools_4.2.3       digest_0.6.33     viridisLite_0.4.2 evaluate_1.0.3
##  [9] lifecycle_1.0.4   tibble_3.2.1      gtable_0.3.6      pkgconfig_2.0.3
## [13] rlang_1.1.2       cli_3.6.2         rstudioapi_0.17.1 yaml_2.3.8
## [17] xfun_0.41         fastmap_1.2.0     xml2_1.3.6        withr_3.0.2
## [21] dplyr_1.1.4       stringr_1.5.1     systemfonts_1.0.5 generics_0.1.3
## [25] vctrs_0.6.5       grid_4.2.3        tidyselect_1.2.0  svglite_2.1.3
## [29] glue_1.6.2        R6_2.6.1          rmarkdown_2.29    magrittr_2.0.3
## [33] scales_1.3.0      promises_1.2.1    htmltools_0.5.7   mime_0.12
## [37] colorspace_2.1-0  xtable_1.8-4      httpuv_1.6.13     stringi_1.8.3
## [41] munsell_0.5.1
```